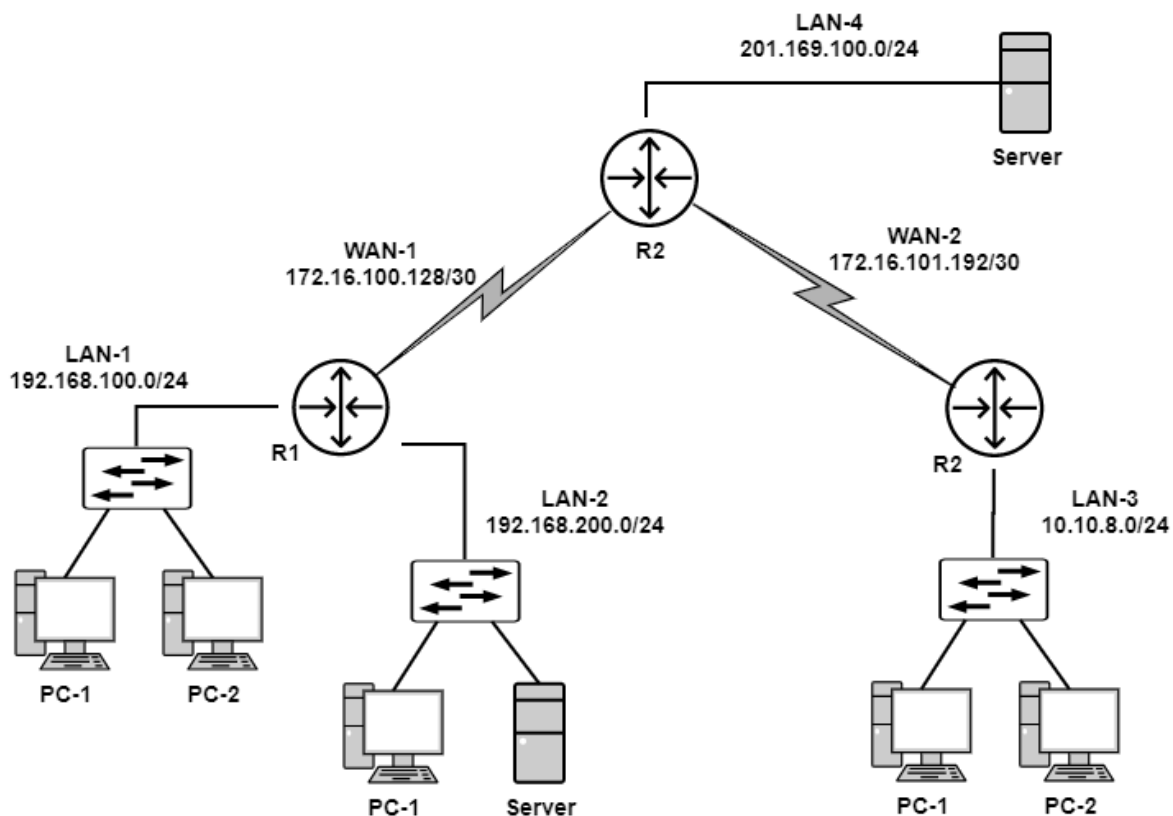


Lab-5 – Internetworking: Dynamic Routing (RIP and OSPF) Concept

Objectives

- Types of Dynamic Routing Protocols
- Difference between RIP and OSPF
- Configure and verify Dynamic Routing (i.e. RIP and OSPF)
- Verify the Routing Table and analysis.

Routing Topology



Part-A: Dynamic Routing: RIP

Part-1: Background: RIP (Routing Information Protocol)

In this activity, you will configure a simple dynamic routing protocol named **Routing Information Protocol (RIP)**. RIP is a relatively old but still commonly used interior gateway protocol created for use in small, homogeneous networks. It is a classical **distance-vector** routing protocol.

Part-2: Configuring Routing Information Protocol (RIP)

For this router to participate in a dynamic routing using a dynamic routing protocol like RIP, you'll need to enable a routing protocol and advertise **the directly connected networks** that want advertised.

Part-3: Routing Table of the routers

R1 # show ip route

Part-4: Write the Connect network of Router R1, R2 and R3

Router R1 Directly Connected Networks

192.168.100.0/24

192.168.200.0/24

172.16.100.128/30

Router R2 Directly Connected Networks

Router R3 Directly Connected Networks

Part-5: Enable RIP and Create RIP Table

To enable a dynamic routing protocol, enter global configuration mode and use the router command. Enter “**router ?**” at the global configuration prompt to see a list of available routing protocols on your router. To enable RIP, enter the command `router rip` in global configuration mode. Once you are in routing configuration mode, enter the network address for **each directly connected network**,

Command Syntax,

Router (config) # router rip

Router (config-router) # network <Connected Network>

Router (config-router) # exit

Command:

R1 (config) # router rip

R1 (config-router) # network 192.168.100.0

R1 (config-router) # network 192.168.200.0

R1 (config-router) # network 172.16.100.128

R1 (config-router) # exit

R1 (config) #

Part-6: Enable RIP version 2 for Classless Network Address

```
R1 (config) # router rip
R1 (config-router) # rip version 2
R1 (config-router) # exit
```

Note: Now you configure the R2 and R3 Router.

Part-7: Verify Routing table and Test the network and Save to the NVRAM

```
R1 # show ip route
R2 # show ip route
R3 # show ip route
```

Check your routing table for entries that are preceded by a capital letter "**R**" to ensure that you are receiving routing updates using RIP. Use show ip route to see the routing table. Ensure that both routers configured so that you can receive his updates. No updates, no ping

When you are finished with the routing configuration, return to privileged EXEC mode and save the current configuration to NVRAM.

```
R1#copy run start
```

Part-B: Dynamic Routing: OSPF

Part-1: Background: OSPF (Open Shortest Path First)

Open Shortest Path First (OSPF) is a routing protocol for Internet Protocol (IP) networks. It uses a link state routing (LSR) algorithm and falls into the group of interior gateway protocols (IGPs), operating within a single autonomous system (AS).

Part-2: Configuring OSPF (Open Shortest Path First)

For OSPF, you'll need to enable a routing protocol and advertise **the directly connected networks** that want advertised. We also need an **Area number, OSPF Process ID** and **Router ID**.

Part-3: OSPF Process ID, Router ID and Area

The Process ID is locally significant and is needed to identify a unique instance of an OSPF database. Process ID number can be from 0 to 65535.

The router ID is a 32-bit number that identifies the OSPF router. Just like IPv4 addresses, it's an 8-bit decimal number separated by "." (dot)" and then write four of them in a row. **The router ID is the name of the OSPF router**, so to speak, and a unique router ID is always required for OSPF processing. The router ID recognizes the neighbor. The LSA is also marked with the router ID of the generated router.

OSPF has two types, one is single area OSPF and Multi area OSPF. In this course we cover the single area OSPF. In single area OSPF we use **area 0**.

Part-4: Enable OSPF and Create OSPF Table

Based on **Part-4 of Part-A**, to enable OSPF, enter the command “Router OSPF” with OSPF process ID in global configuration mode. Once you are in routing configuration mode, enter the network address for **each directly connected network**,

Command Syntax,

```
Router (config) # router ospf <Process ID>
Router (config-router) # router-id <Router ID>
Router (config-router) # network <Connected Network> <Wildcard Mask>
<Area>
Router (config-router) # exit
```

Command:

```
R1 (config) # router ospf 100
R1 (config-router) # router-id 100.100.100.1
R1 (config-router) # network 192.168.100.0 0.0.0.255 area 0
R1 (config-router) # network 192.168.200.0 0.0.0.255 area 0
R1 (config-router) # network 172.16.100.128 0.0.0.3 area 0
R1 (config-router) # exit
R1 (config) #
```

Note: Now you configure the R2 and R3 Router.

Part-5: Verify Routing table and Test the network

Follow the **Part-7 of Part-A**

Lab-6 (Part-1) Dynamic Host Configuration Protocol (DHCP)

Objectives

- To learn about Dynamic Host Configuration Protocol (DHCP): why and how used?
- To build an internetwork and configure DHCP using Packet Tracer

Part-1: Background

Dynamic Host Control Protocol (DHCP) enables you to automatically assign reusable IP addresses to **DHCP clients**. The **DHCP Server** feature is a full **DHCP** server implementation that assigns and manages IP addresses from specified address pools within the router to DHCP clients. **Figure 1** shows the basic steps that occur when a DHCP client requests an IP address from a DHCP server. The client, Host A, sends a **DHCPDISCOVER** broadcast message to locate a DHCP Server. A DHCP server offers configuration parameters (such as an IP address, a MAC address, a domain name, and a lease for the IP address) to the client in a **DHCPOFFER** unicast message.

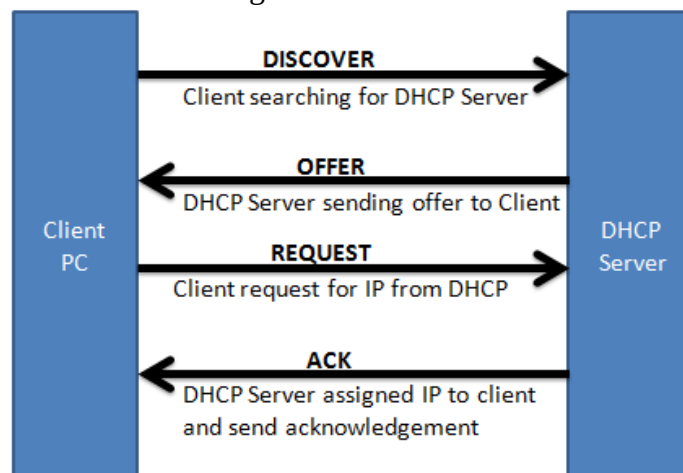


Figure 1: DHCP client/server mode

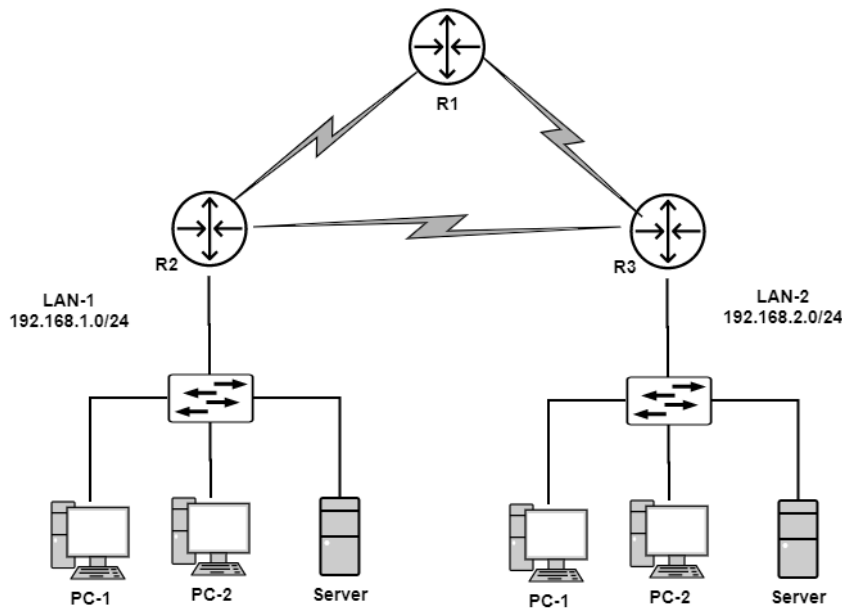
Benefits of use DHCP

Reduced Internet access costs: Using automatic IP address assignment at each remote site substantially reduces Internet access costs. Static IP addresses are considerably more expensive to purchase than are automatically allocated IP addresses.

Reduced client configuration tasks and costs: Because DHCP is easy to configure, it minimizes operational overhead and costs associated with device configuration tasks and eases deployment by nontechnical users.

Centralized management: Because the DHCP server maintains configurations for several subnets, an administrator only needs to update a single, central server when configuration parameters change.

Part-2: DHCP Server Configuration



Part-2(A): DHCP IP POOL

We configure DHCP in LAN-1 Network and choose any name for IP POOL. But as a network professional we maintain some DHCP Pool create rules. We take the 1st the Router name and then the interface name. Example, LAN-1 network DHCP POOL name: **R2G0/0**

```
R2 (config)# ip dhcp pool R2G0/0
```

Specify the subnet to use when assigning IP addresses. DHCP pools automatically associate with an interface based on the network statement.

```
R2 (config-dhcp)# network 192.168.1.0 255.255.255.0
```

Configure the **default router** and **domain name** server for the network. Clients receive these settings via DHCP, along with an IP address.

```
R2 (config-dhcp)# default-router 192.168.1.1
```

```
R2 (config-dhcp)# dns-server 8.8.8.8
```

```
R2 (config-dhcp)#exit
```

Part-2(B): Exclude statically assigned addresses

The DHCP server assumes that all IP addresses in a DHCP address pool subnet are available for assigning to DHCP clients. You must specify the IP addresses that the DHCP server should not assign to clients. These IP addresses are usually static addresses reserved for the router interface, switch management IP address, servers, and local network printer. The `ip dhcp excluded-address` command prevents the router from assigning IP addresses within the configured range. The following commands exclude the first 10 IP addresses from each pool for the LANs attached to R1. These addresses will not be assigned to any DHCP clients.

```
R2(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

Part-2(C): Test the DHCP configuration

- Select **DHCP** option to set PC-1 and PC-2 to get dynamic IP from the **DHCP Server**.
- Configure the **Server** with the **static** address **192.168.1.5**.

It is assumed that the server requires a **fixed IP address** in order to allow access from any PC in the LAN. Configure the server address manually. Ping from the PC2 and PC3 to router ports and server host.

Part-3: Configure other Network

Configure LAN-2 Network DHCP

Lab-6 (Part-2): Introduction to Wireshark

Objectives

- To learn about **Wireshark**.
- Packet Capture & Traffic Analysis
- Analyze the Packets using Wireshark

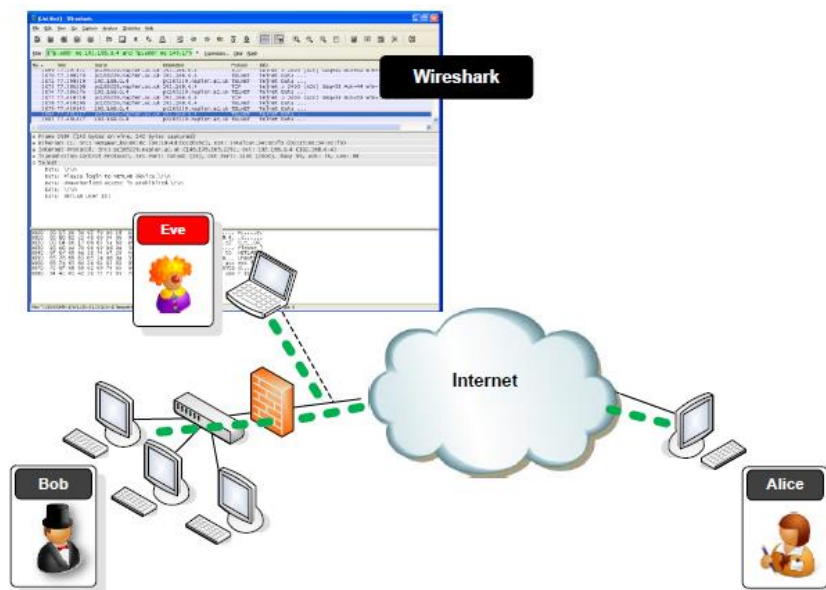
Part-1: Wireshark

Wireshark is a free and open-source packet analyzer. It is used for network troubleshooting, analysis, software and communications protocol development, and education. Develop by The Wireshark team. Website, www.wireshark.org

Wireshark is cross-platform, using the Qt widget toolkit in current releases to implement its user interface, and using pcap to capture packets; it runs on Linux, macOS, BSD, Solaris, some other Unix-like operating systems, and Microsoft Windows. There is also a terminal-based (non-GUI) version called TShark. Wireshark, and the other programs distributed with it such as TShark, are free software, released under the terms of the GNU General Public License.

Part-2: Packet Capture/Package Sniffing

A packet sniffer is an application which can capture and analyses network traffic which is passing through a system's Network Interface Card (NIC). The sniffer sets the card to promiscuous mode which means all traffic is read, whether it is addressed to that machine or not. The figure below shows an attacker sniffing packets from the network, and the Wireshark packet sniffer/analyzer (formerly known as ethereal).



Part-3: Packet Analysis

Wireshark is an open source cross-platform packet capture and analysis tool, with versions for Windows and Linux. The GUI window gives a detailed breakdown of the network protocol stack for each packet, colorizing packet details based on protocol, as well as having functionality to filter and search the traffic, and pick out TCP streams. Wireshark can also save packet data to files for offline analysis and export/import packet captures to/from other tools. Statistics can also be generated for packet capture files.